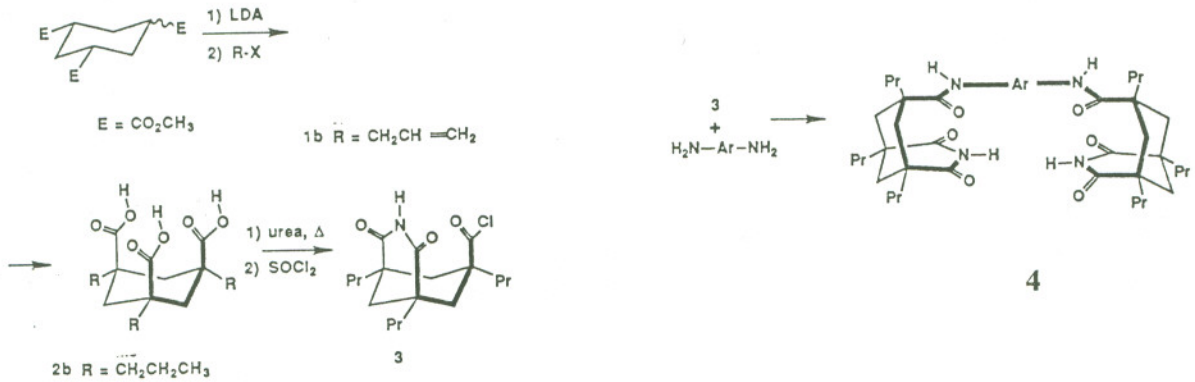
Organic Chemistry
Cumulative Exam
September 15, 1990

Understanding the factors involved in *molecular recognition* is an important area of current interest in bioorganic chemistry. Julius Rebek, Jr. and his colleagues at MIT have made important contributions to this area including a recent article in *J. Am. Chem. Soc.* **1990**, *112*, 6144-6145.

Receptors for substituted lactams were synthesized according to **Scheme I.**

Scheme I



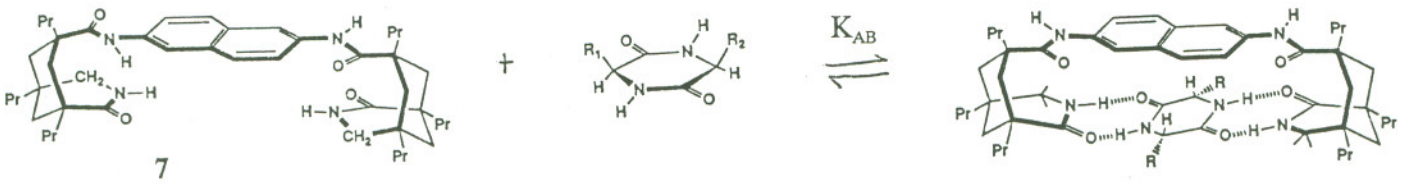
a) Indicate the structure of **1b** if R-X = $CH_2{=}CHCH_2$-X.

b) Indicate the reagents needed to convert **1b** to **2b.**

c) Describe how reaction of **3** with H_2N-Ar-NH_2 yields **4.**

Compound **4** is converted in several steps to the final molecular receptor **7** that binds diketopiperizines with association constants K_{AB} as large as K_{AB} = 82,000 M^{-1} at 25 °C in $CDCl_3$.

Scheme II



d) Explain why these association constants are so large and calculate the free energy difference between reactant and products for **Scheme II.** *Hint*: a useful relation is $\Delta G° = -RT\ln K$ with R = 1.99 cal K^{-1} mol^{-1}.